AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS 2021

Biology

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

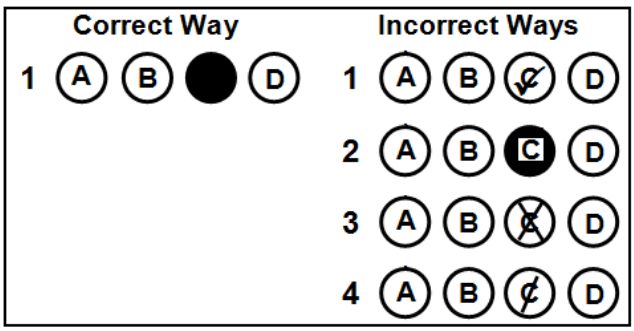
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.

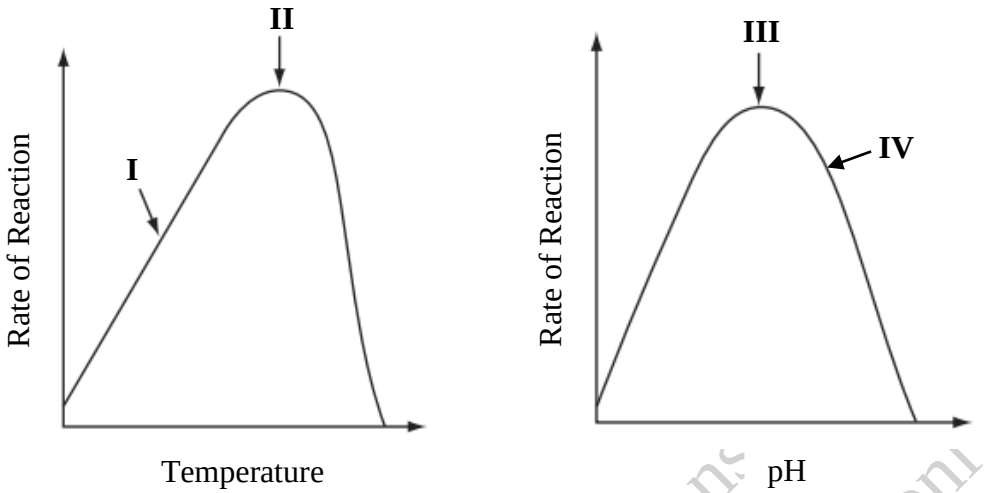


Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. The given graphs show the effects of temperature and pH on enzyme activity.



Which of the given statements CORRECTLY matches the point on the graphs with the description of enzyme activity?

A. At point I, hydrogen bonds are formed between enzyme and substrate.
B. At point II, kinetic energy of enzyme and substrate is the highest.
C. At point III, peptide bonds in the enzyme begin to break.
D. At point IV, the enzyme is completely denatured.

2. The MAIN purpose of feedback inhibition in a metabolic pathway is to

A. activate the enzymes.
B. alter the rate of reactions.
C. minimise the energy efficiency.
D. regulate the amount of product.

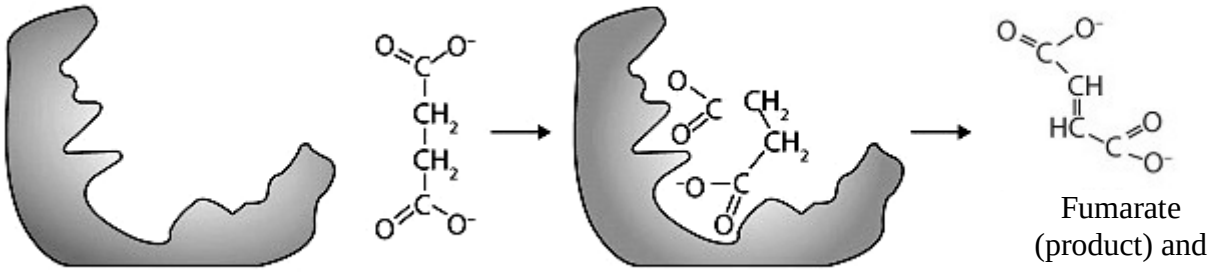
3. In contrast to coenzymes, prosthetic groups are

A. derived from vitamins.
B. composed of amino acids.
C. loosely attached to enzymes.
D. covalently attached to enzymes.

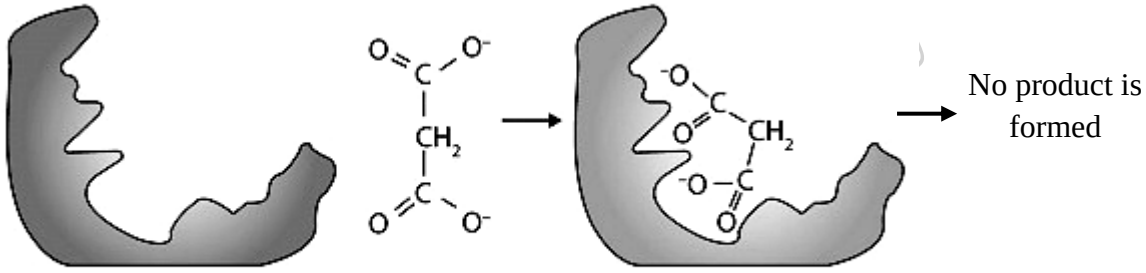
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4. Consider the given metabolic reactions **I** and **II**.

Metabolic reaction I



Metabolic reaction II



In the metabolic reaction **II**, product is NOT formed because malonate acts as a/ an

A. allosteric inhibitor.
B. irreversible inhibitor.
C. competitive inhibitor.
D. non-competitive inhibitor.

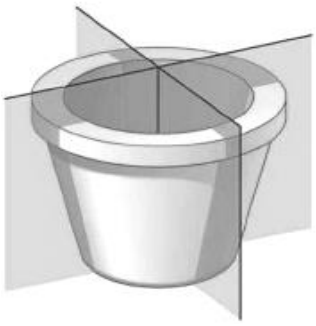
5. The option that CORRECTLY identifies the class, type of coelom and organisation of germinal layers in leech is

	Class	**Type of Coelom**	**Organisation**
A	hirudinea	coelomate	triploblastic
B	polychaeta	coelomate	triploblastic
C	oligochaeta	pseudocoelomate	diploblastic
D	hirudinea	acoelomate	diploblastic

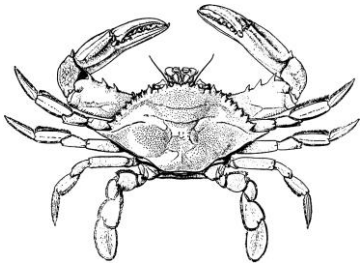
6. The characteristic of prototherians that provides an evidence of evolution of mammals from reptiles is that they have

A. webbed toes.
B. thick fur on their bodies.
C. mammary glands to feed their new-borns.
D. the same opening for excretion and reproduction.

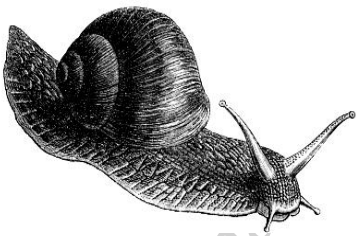
7. The given diagram represents a type of body symmetry.



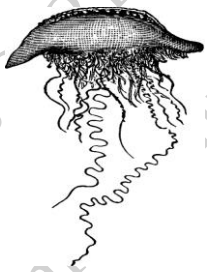
Which of the following animals has this type of body symmetry as shown in the given diagram?



A



B



C



D

8. The CORRECT feature of ostium and osculum in the body of sponges is

	Ostium	Osculum
A	single cavity inside the body	pore by which the water leaves the body
B	pore through which water enters the body	pore by which the water leaves the body
C	pore through which water enters the body	gelatinous mesenchyme which may contain cells
D	single cavity inside the body	gelatinous mesenchyme which may contain cells

9. The example of an animal which has a complete digestive system is

A. jelly fish.
B. tapeworm.
C. liver fluke.
D. round worm.

10. In contrast to amphibians, reptiles

A. have separate sexes.
B. respire through lungs.
C. lay eggs with soft and leathery shells.
D. cannot control their body temperature.

11. In the classification of kingdom animalia, poriferans are included in subkingdom parazoa because they

    A. are found in water.
    B. have the ability to regenerate.
    C. demonstrate division of labour.
    D. show cellular grade of organisation.

12. The option that CORRECTLY identifies the type of endoskeleton and existence of jaws in a lamprey fish is

	Type of Endoskeleton	**Jaws**
A	cartilaginous	present
B	bony	present
C	cartilaginous	absent
D	bony	absent

13. The type of locomotion shown by bacteria towards carbon and energy sources is

    A. positive chemotaxis.
    B. negative chemotaxis.
    C. positive magnetotaxis.
    D. negative magnetotaxis.

14. *Escherichia coli* is a facultative anaerobic bacterium because it can

    A. not survive in the presence of oxygen.
    B. utilise low concentration of oxygen only.
    C. survive without oxygen, but uses oxygen if it is present.
    D. neither use oxygen for its growth nor can be harmed by it.

15. The pathogenic protozoan species that belongs to class apicomplexa is

    A. entamoeba.
    B. paramecium.
    C. plasmodium.
    D. trypanosoma.

16. In contrast to plants and animals, protists do NOT

    A. have cell walls.
    B. reproduce asexually.
    C. develop from an embryo.
    D. possess membrane bounded organelles.

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17. Given are some of the groups of protozoa.

    I. Actinopods
    II. Apicomplexans
    III. Foraminiferans

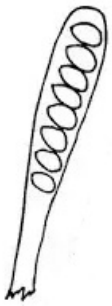
    Pseudopodial mode of locomotion is found in group(s)

    A. I only.
    B. II only.
    C. I and III.
    D. II and III.

18. The presence of lichen in an area indicates the

    A. absence of air pollution.
    B. nutrient content of the area.
    C. rich biodiversity of the area.
    D. beginning of ecosystem degradation.

19. The given sexual reproductive structure of fungi belongs to phylum



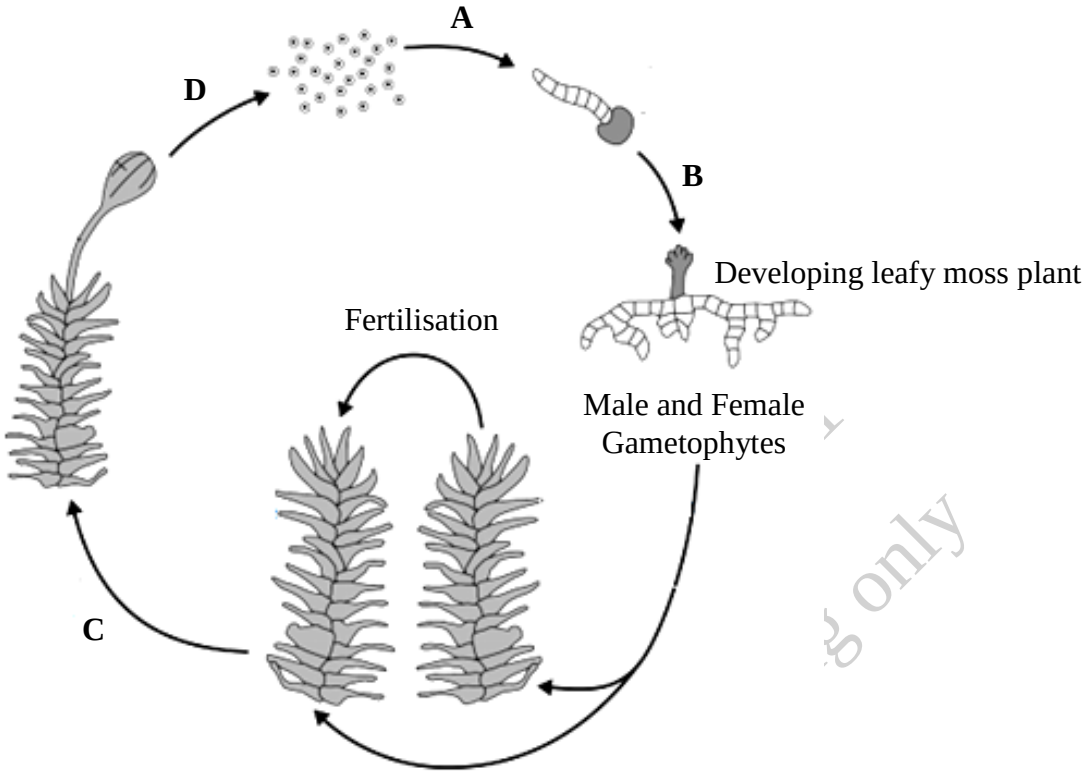
    A. ascomycota.
    B. zygomycota.
    C. basidiomycota.
    D. deuteromycota.

20. The option that CORRECTLY matches the fungal hypha and zygote with its respective ploidy is

	Fungal Hypha and Zygote	**Ploidy**
A	dikaryotic hypha	polyploid
B	zygote	polyploid
C	dikaryotic hypha	haploid
D	zygote	haploid

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21. In the given diagrammatic life cycle of moss plant, the stage at which meiosis occurs is



22. The role of the endosperm tissue of seeds is to

A. protect the female gametes.
B. produce the female gametes.
C. secrete enzymes for seed growth.
D. provide nutrition to the developing embryo.

23. An example of a monocot leaf is

			
A	B	C	D

24. The significance of heterospory in plants is that it

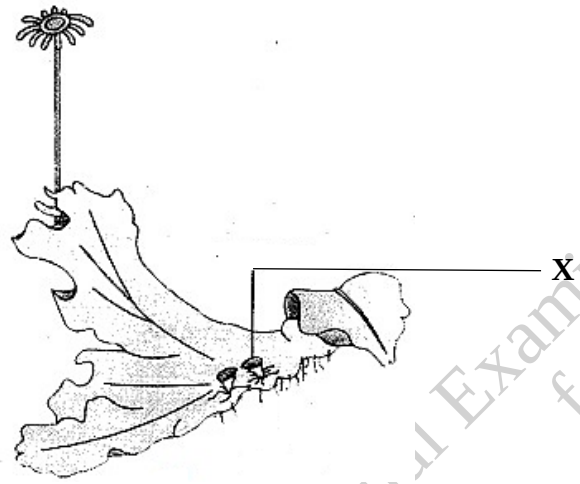
A. helps in seed formation.
B. ensures the development of motile sperms.
C. provides an efficient way of spore dispersion.
D. helps in the production of independent gametophytes.

PLEASE TURN OVER THE PAGE

25. In spermatophytes (seeded plants), the ploidy of embryo and the generation in which the embryo is produced is

	Ploidy of Embryo	Generation
A	diploid	sporophyte
B	haploid	sporophyte
C	diploid	gametophyte
D	haploid	gametophyte

26. The given diagram shows a Marchantia thallus (liverwort).



The function of structure **X** is to

A. provide protection.
B. provide nourishment.
C. help in gametes production.
D. help in vegetative propagation.

27. In cellular respiration, the reaction that does NOT occur during the formation of acetyl CoA is

A. decarboxylation.
B. reduction of NAD^+.
C. formation of ATP.
D. formation of acetyl group.

28. Photosynthesis in C3 plants is reduced in the presence of excessive

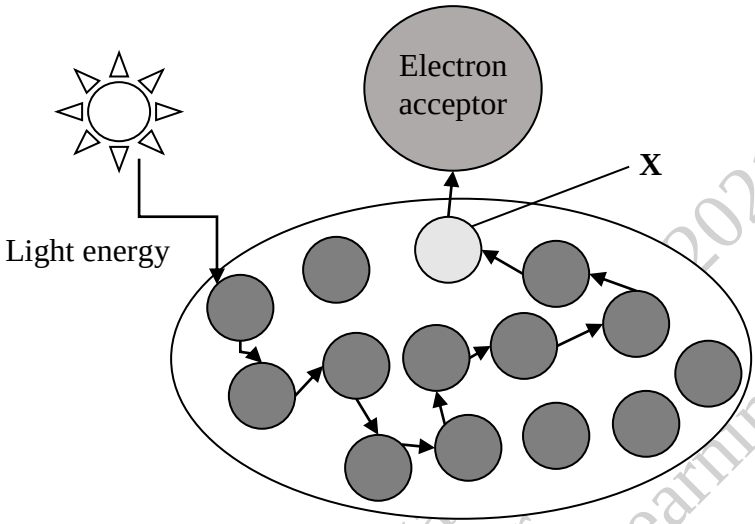
A. oxygen.
B. rubisco.
C. carbon dioxide.
D. glyceraldehyde-3-phosphate.

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29. Oxygen is required in aerobic respiration for

A. the reduction of hydrogen carriers.
B. the oxidation of glyceraldehyde 3-phosphate.
C. accepting electrons at the end of the electron transport chain.
D. maintaining an oxygen concentration gradient in mitochondria.

30. The given diagram shows a photosystem found in thylakoid membranes.

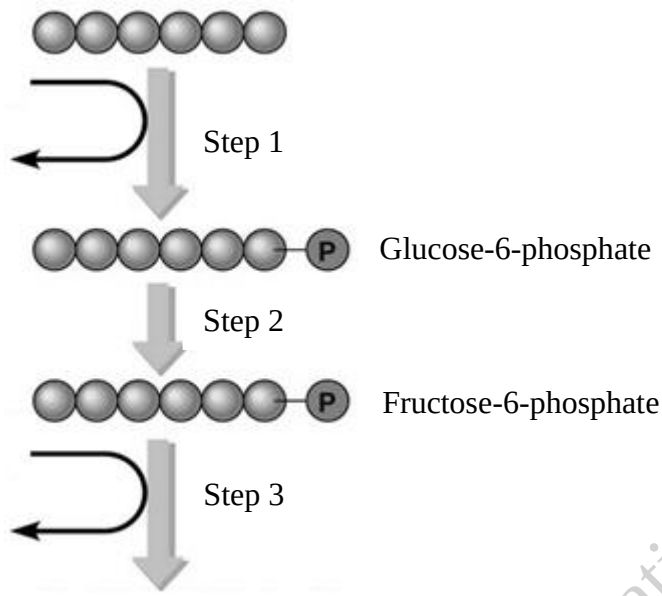


The CORRECT description of structure **X** is that it

A. is a light harvesting complex.
B. passes on energy to other pigment molecules.
C. has one or more molecules of chlorophyll a.
D. contains molecules of carotenoids and xanthophyll.

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31. The given diagram shows some of the reactions that occur in the preparatory phase of glycolysis.



The conversion that takes place at step 1 and 3 is

A. ATP → ADP.
B. ADP → ATP.
C. NAD^+→ NADH.
D. NADH → NAD^+.

32. An animal in which food is completely digested in the gut is

A. hydra.
B. amoeba.
C. planaria.
D. cockroach.

33. In the gastric glands of the human digestive system, the secretion of parietal cell and zymogen cell respectively is

	Parietal Cell	Zymogen Cell
A	gastrin	hydrochloric acid
B	hydrochloric acid	gastrin
C	hydrochloric acid	pepsinogen
D	gastrin	pepsinogen

34. In the human digestive system, the stomach empties chyme into the duodenum when the

A. pyloric sphincter is relaxed.
B. cardiac sphincter is relaxed.
C. pyloric sphincter is contracted.
D. cardiac sphincter is contracted.

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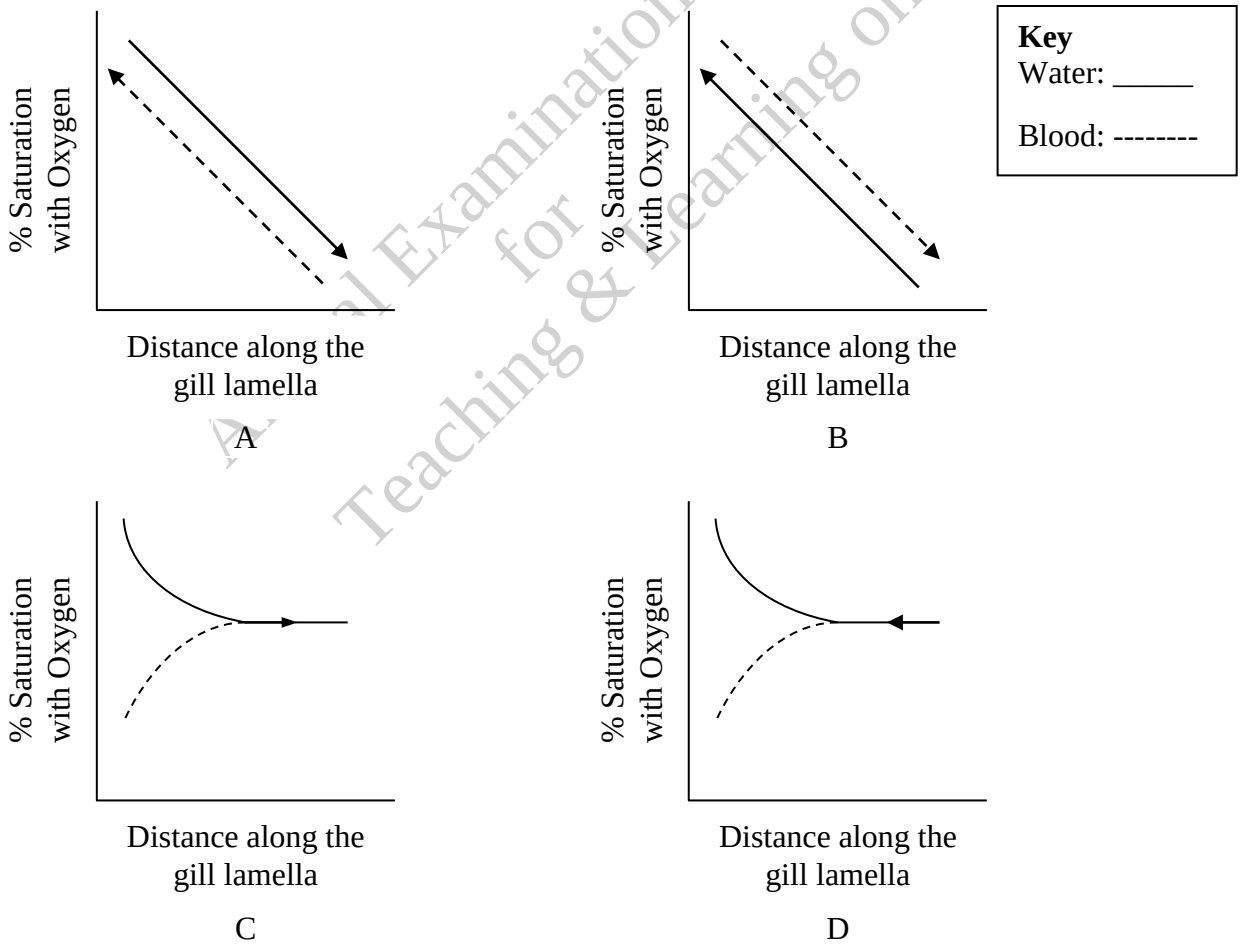
35. An example of endoparasite in animals is

A. lice.
B. ticks.
C. leech.
D. tapeworm.

36. The organs of the human digestive system other than the small intestine, that secrete protein-digesting enzymes are

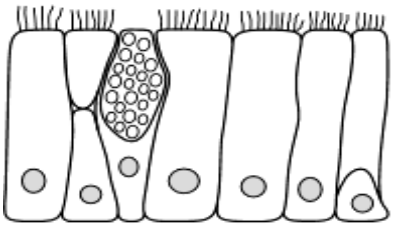
A. stomach and liver.
B. liver and pancreas.
C. stomach and pancreas.
D. liver and large intestine.

37. The graph that CORRECTLY interprets the counter current flow with respect to oxygen concentration as water flows across the gill lamellae of fish is



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38. The given diagram shows a type of epithelium.



The parts of respiratory system lined by the given type of epithelium are

A. pharynx and larynx.
B. pharynx and trachea.
C. trachea and bronchus.
D. larynx and bronchioles.

39. The factor(s) that could cause emphysema in humans is/ are

I. urban smog
II. bacterial infection
III. tobacco smoke

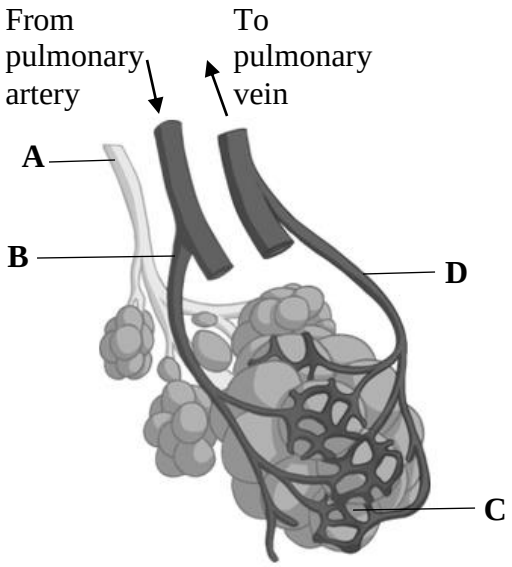
A. I only.
B. II only.
C. I and III.
D. II and III.

40. The type of respiration which does NOT involve blood is the

A. pulmonary respiration in frogs.
B. tracheal respiration in cockroach.
C. cutaneous respiration in earthworms.
D. bucco-pharyngeal respiration in frogs.

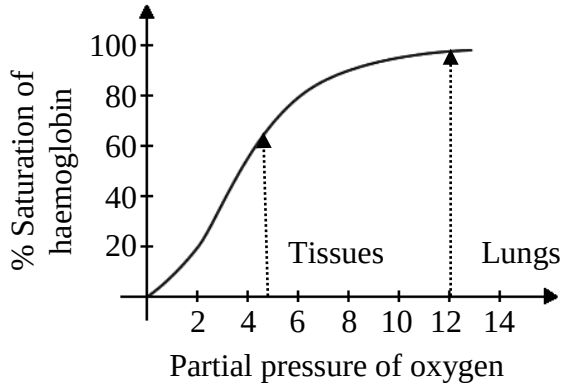
41. The given diagram shows air sacs and associated blood vessels in the human lung.

The concentration of carbon dioxide is the highest in the part labelled as



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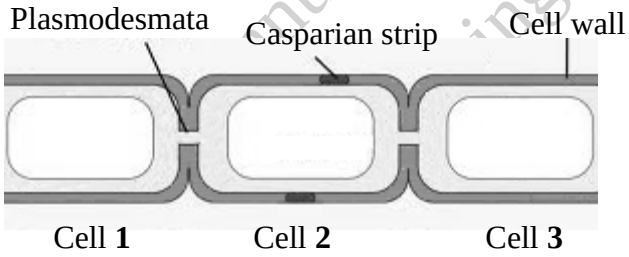
42. The given graph shows the haemoglobin saturation curve.



The relative oxygen concentration in body tissues and lungs as presented in the graph is

	Relative Oxygen Concentration in	
	Body Tissues	**Lungs**
A	low	high
B	low	low
C	high	high
D	high	low

43. The given diagram shows three plant cells arranged side by side.



The movement of water molecules from cell **1** to cell **3** would discontinue in case of

A. apoplast pathway.
B. symplast pathway.
C. vacuolar pathway.
D. trans-membrane pathway.

44. The large arteries close to the heart have a thick elastic layer in their walls to

A. prevent collapse of these blood vessels.
B. reduce friction within these blood vessels.
C. anchor these vessels with surrounding tissues.
D. prevent too much pressure exerted on their walls.

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45. The rate of blood flow is the slowest in capillaries than other blood vessels because the

A. capillaries lack valves.
B. capillaries have one cell thick walls.
C. resistance in blood flow is negligible in capillaries.
D. total cross-sectional area of the capillaries is greater than other vessels.

46. A person has ventricular septal defect-a hole in the wall (septum) that separates the two lower chambers (ventricles) of the heart.

What kind of problems will this person encounter?

I. More oxygen poor blood will be contained in the right atrium.
II. Volume of the blood circulating through the lungs will be higher.
III. Less oxygen will be delivered to the body tissues.

A. I only
B. III only
C. I and II
D. II and III

47. When blood pressure is the highest in the aorta, the atrioventricular and semilunar valves would

	Atrioventricular Valve	**Semilunar Valve**
A	open	close
B	close	open
C	close	close
D	open	open

48. The blood pressure of a person is 170/105 mm Hg.

Which of the following is the BEST interpretation of this reading?

A. Normal
B. Hypotension
C. Mild hypertension
D. Hypertension requiring treatment

49. The reason for an increase in the number of white blood cells in people with leukaemia is that more

A. memory cells are produced in the plasma.
B. lymphocytes are produced in the bone marrow.
C. T-lymphocytes are required to stimulate the immune system.
D. antibodies are required to increase the production of B-lymphocytes.

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50. An individual bitten by a rabid dog can be treated by an injection of human rabies antibodies.

The type of immunity provided by this treatment is

A. natural active.
B. natural passive.
C. artificial active.
D. artificial passive.

ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

51. Saqib performed the following steps in sequence as a part of an experiment.

- Added equal amounts of starch solution into two test tubes, **I** and **II**
- Added salivary amylase to test tube **I** and boiled amylase solution to test tube **II**
- Shook the test tubes and left them in a water bath for 10 minutes at 37°C
- Added a few drops of iodine solution to each test tube

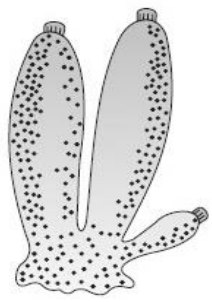
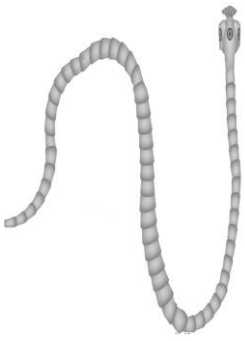
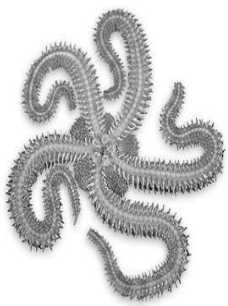
He observed that the contents of test tube **I** went reddish yellow in colour, and the test tube **II** maintained its blue colour.

The CORRECT inference that he must have made based on his observations of test tube **I** and test tube **II** is

	Test Tube I	**Test Tube II**
A	amylase hydrolysed the starch to fructose.	starch is not hydrolysed by boiled amylase.
B	starch is not hydrolysed by amylase	boiled amylase hydrolysed the starch to maltose.
C	amylase converts the starch to fructose.	boiled amylase hydrolysed the starch to maltose.
D	amylase hydrolysed the starch to maltose.	starch is not hydrolysed by boiled amylase.

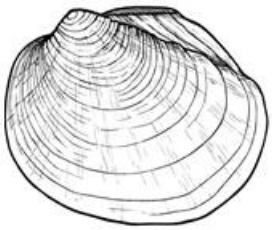
52. A student observed the following four invertebrate specimens in the biology laboratory.

The specimen that has bilateral symmetry is

			
A	B	C	D

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53. In the biology laboratory, a student observed the following specimen.



The identification and distinguishing feature of this specimen is

	Identification	Distinguishing Feature
A	mussel	bivalved shell
B	mussel	soft body
C	slug	bivalved shell
D	slug	soft body

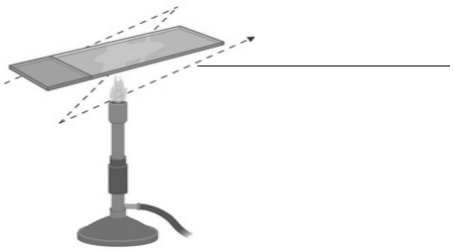
54. Jameel identifies Gram-positive and Gram-negative bacteria by using the Gram staining technique. After the completion of the whole procedure, he observed under the microscope that the Gram-positive bacteria were stained purple (violet) and Gram-negative bacteria were red (pink) in colour.

The reason for this difference in staining is that the cell wall of Gram-positive bacteria contains a layer that retains the purple crystal violet stain.

This layer is composed of

A. lipoprotein.
B. peptidoglycan.
C. cellulose.
D. hemicellulose.

55. Hammad prepared a slide of bacterial smear for Gram staining. He quickly passed the slide over a gentle flame as shown in the given diagram.

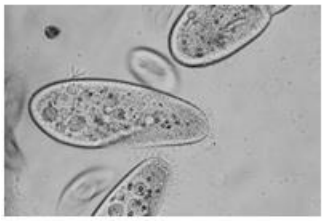


This step is important for Gram staining because it

A. visualises the entire fauna of microbes.
B. decolourises the bacteria for staining properly.
C. stops the further growth of bacteria in the smear.
D. prevents the smear from washing off during staining.

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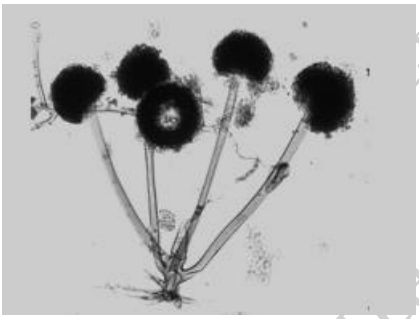
56. Adeel observed the given microscopic view of a permeant slide in the laboratory.



The CORRECT identification and the reason for identification is

	Identification	Reason for Identification
A	euglena	spindle-shape
B	euglena	slipper-shape
C	paramecium	spindle-shape
D	paramecium	slipper-shape

57. A student prepared a temporary slide of the fresh culture provided and examined it under the microscope.



The microscopic view depicts that the unknown material is

A. yeast.
B. rhizopus.
C. mushroom.
D. penicillium.

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58. Given are the images of three flowers.



I **II** **III**

The polypetalous flower(s) in the given images is/ are

A. I only.
B. II only.
C. I and II.
D. II and III.

59. Given is the floral diagram of a flower.



The floral formula for this flower is

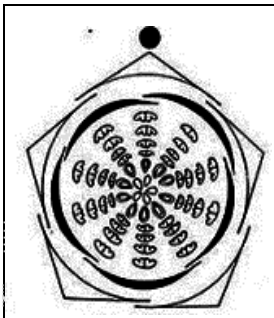
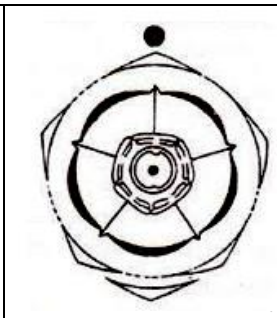
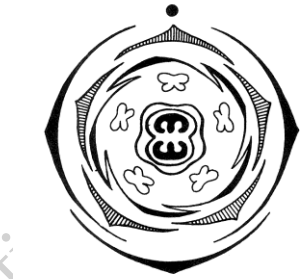
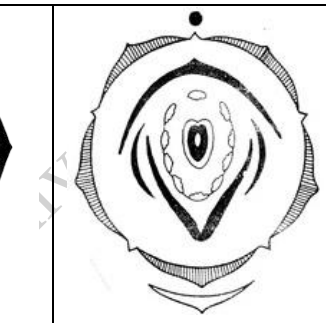
A. ⊕ ,♂, K_5,$C_{1+2+2,}$ A_{9+1}, $\underline{G}_1$
B. ⊥ ,♀ , $K_{(5)}$,$C_{1+2+(2),}$ $A_{(9)+1}$, $\underline{G}_1$
C. ⊥, ♂,$K_{(5)}$,$C_{1+2+(2),}$ $A_{(9)+1}$, $\underline{G}_1$
D. ⊕ ,♀, $K_{(5)}$,$C_{1+2+(2),}$ $A_{(9)+1}$, $\underline{G}_1$

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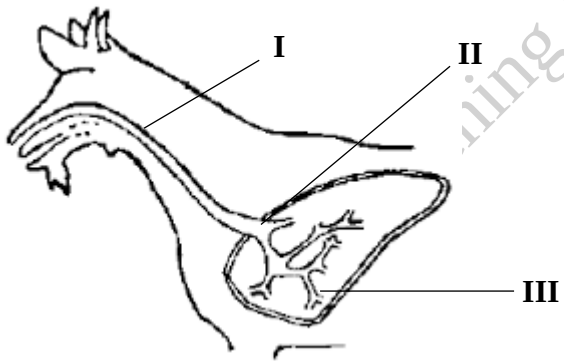
60. Four students (A, B, C, D) drew the floral diagram for the given flower.



The CORRECT floral diagram is drawn by student

			
A	B	C	D

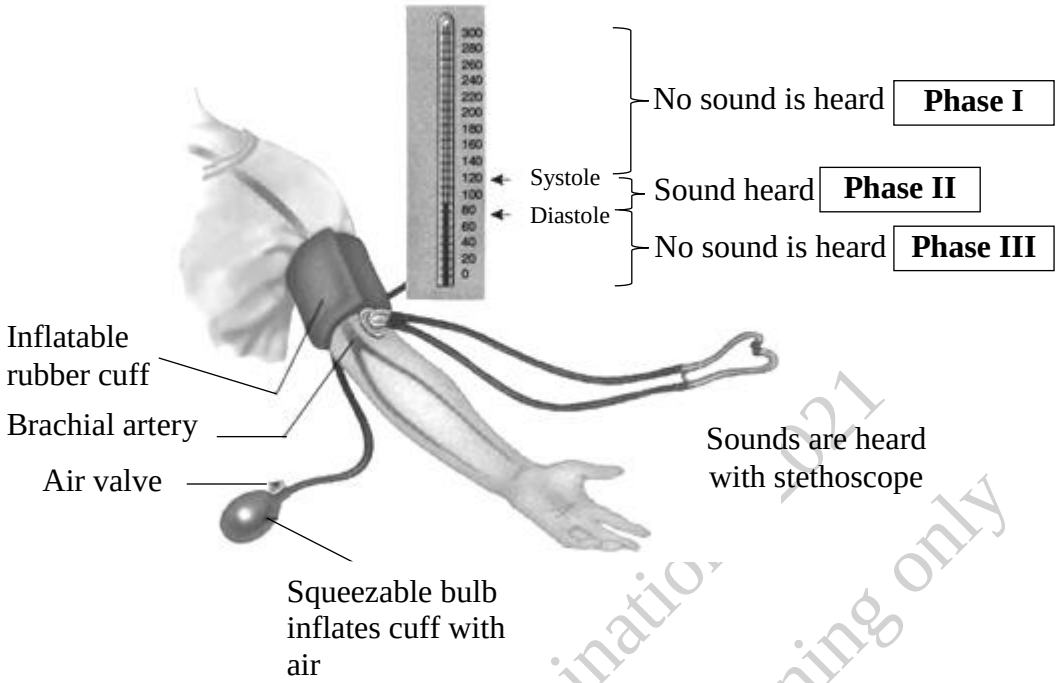
61. Given is the diagram of the respiratory system of a goat.



The part of the air passage which lack(s) cartilage is

A. I only.
B. I and II.
C. III only.
D. II and III.

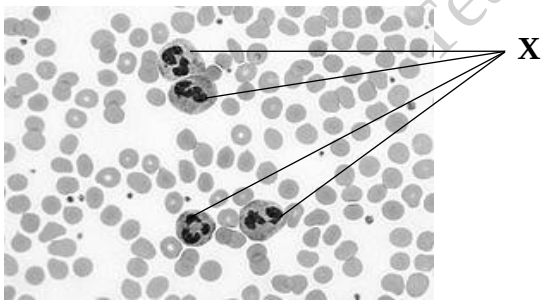
62. The given diagram shows the method of measuring the blood pressure of a healthy person.



At phase **III**, sound is not heard because the brachial artery

A. is closed.
B. is opened.
C. is bearing the greatest pressure.
D. shows a continuous change in the diameter of lumen.

63. The given microscopic view of the human blood smear was observed in a laboratory.



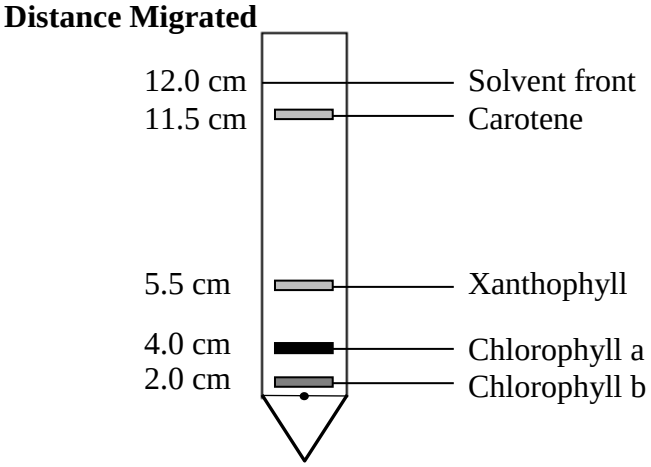
The option that identifies the blood cells labelled as **X** and the reason for their identification is

	Blood Cell X	**Reason for Identification**
A	neutrophils	multilobed nucleus
B	basophils	multilobed nucleus
C	neutrophils	bilobed nucleus
D	basophils	bilobed nucleus

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Use the given diagram to answer Q.64 and Q.65.

The given diagram shows the plant pigments separated on a paper chromatogram.



64. By using the given chromatogram, the Rf value of 'chlorophyll a' is

A. 0.45
B. 0.95
C. 0.33
D. 0.16

65. The LEAST soluble pigment is identified as

A. carotene.
B. xanthophyll.
C. chlorophyll a.
D. chlorophyll b.

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